Gokulakrishnan M

AI Engineer | ML Engineer | Data Scientist

in/gokulkrishnan-m | Portfolio | gokulakrishnan.m0999@gmail.com | +91-7639273373

Education

M.Tech, Biomedical Engineering	*IIT Bombay*	**CPI: 9.45**	2021 - 2023
B.E, Biomedical Engineering	*CEG, Anna University*	**GPA: 7.6**	2017 - 2021

Industry Experience

AI Engineer | *Thisai (Contract)* Jan 2026 - Oct 2026

- **Local LLM Evaluation:** Benchmarked self-hosted Llama, Qwen, and GPT-OSS (3B-20B) with prompt and inference tuning inside an LLM pipeline that verifies 14k+ MCQs and generates pedagogical metadata.
- **Computer Vision & GenAI:** Architected an OpenCV + Vision-LLM OCR pipeline on Django/Celery to grade scanned answer sheets; tuned and integrated a Gemma 3 4B RAG system to auto-generate personalized exams.

Data Science Associate | *StepChange Inc* May 2024 - Nov 2025

- **GenAI & LLM Development:** Engineered a LangChain-driven RAG pipeline using **Azure OpenAI** APIs to automate policy report generation and predict qualitative outcomes from historical datasets.
- **AWS Serverless Automation:** Developed an event-driven **AWS Lambda** architecture to trigger data ingestion and real-time metric computation, cutting turnaround from months of manual processing to hours.
- **SaaS Platform:** Optimized high-throughput sustainability metric computation for a B2B carbon accounting platform and designed scalable multi-table data ingestion frameworks built in Java (Spring Boot) and SQL.

Data Analyst | *StepChange Inc* Jul 2023 - Apr 2024

- **Geospatial & Emission Modelling:** Developed algorithms integrating **Google Maps APIs** and Geopy to compute travel distances and emissions in compliance with global Carbon Accounting Financials standards.
- **Data Engineering & Migration:** Migrated **14 TB** of enterprise data across a multi-region Microsoft Fabric tenancy; Engineered and deployed APIs to convert NSE XBRL financial filings into structured data.
- **Analytics Dashboards:** Prototyped interactive **Power BI** dashboards on Azure Storage and Fabric Dataflows

Projects

Agentic AI & **LLM Prompting** | AI-integrated job-hunt toolkit featuring ATS resume scoring, AI-driven analysis, Automated JD parsing, and tracking. Story to comic strip generator agent using MCP tools and Gemini LLM. 2026

Medical Image Segmentation | Engineered a deep supervised UNet++ architecture to extract liver tumors on medical CT scans with custom Dice-Cross Entropy loss, achieving **94% IoU**. 2022

Deep Learning Classification | Implemented **ResNet-18** transfer learning for classification; Built MLPs and Perceptrons from scratch to benchmark activation functions on the MNIST dataset. 2022

ML Hyperparameter Tuning | Compared and fine-tuned multiple ML models (SVM, Logistic Regression, Random Forest) to evaluate hyperparameter sensitivity; Developed K-Means clustering for thermal image segmentation. 2021

Position of Responsibility

Student Companion | Mentored 5 students on academic and personal development for an academic year. 2022 - 2023

Social Media Manager | Handled college official cultural's team digital outreach & 40+ members for 2+ years. 2018 - 2021

Awards & Achievements

GATE Distinction | Secured **AIR 132** in GATE 2021 (Biomedical Engineering) among 1684 candidates. 2021

Cultural Excellence | Winner of 7+ inter-cultural competitions & guest performer on National Voters' Day. 2018 - 2020

Technical Skills & Certifications

Languages & Frameworks: Python, PyTorch, Flask, Java (Spring Boot), SQL, LangChain, LangGraph, TensorFlow
AI & Cloud: Azure OpenAI, Agentic AI, MCP, RAG, LLM, AWS (Lambda, S3), Microsoft Fabric, Docker, Git, Power BI
Certifications: Databases and SQL for Data Science with Python - IBM | **Coursera**
Python for Data Science, AI & Development - IBM | **Coursera**

Interests

Hobbies: Vibe Coding, Blogging, Quotes & Story writing, Designing-graphic artist, Game development
Interests: Agentic AI, Frontier AI research, NLP, Generative AI, Deep learning, and Data Science.